

Empirical Evaluation & Benchmark Results

Evaluation Mode: Real Empirical Run (Zero Fabrication Guarantee)

Dataset: data/processed/processed_flows.csv (Holdout Split: 20% Stratified)

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1. Multi-Threshold Performance (Supervised Flow Classifier)

The classifier was evaluated across multiple probability cutoffs. In production NDR, we employ a **Precision-First posture (threshold = 0.85)** to prevent alert fatigue while delegating evasive/novel patterns to the Suricata signature and Autoencoder layers.

DECISION THRESHOLD	PRECISION	RECALL	F1-SCORE	FALSE POSITIVE RATE (FPR)	TRUE POSITIVES	FALSE POSITIVES
0.50	0.9998	0.9996	0.9997	0.0004	12234	3
0.70	0.9998	0.9996	0.9997	0.0004	12234	3
0.85	0.9998	0.9995	0.9996	0.0004	12233	3
0.95	0.9999	0.9990	0.9995	0.0001	12227	1

2. Per-Category Classification Breakdown (Threshold = 0.85)

ATTACK CATEGORY	PRECISION	RECALL	F1-SCORE	SUPPORT
BENIGN	0.9994	0.9996	0.9995	7763
DDOS_FLOOD	0.9998	0.9996	0.9997	12239
macro avg	0.9996	0.9996	0.9996	20002
weighted avg	0.9996	0.9996	0.9996	20002

3. Unsupervised Autoencoder Anomaly Layer

Trained strictly on benign traffic flows with reconstruction error threshold calibrated at **1865.197388**.

Metric	Measured Value
Anomaly Precision	0.0000
Anomaly Recall	0.0000
Anomaly F1-Score	0.0000
Anomaly FPR	0.0116
True Negatives	7673
False Positives	90